

Comprehensive Exploratory Data Analysis, Feature Engineering & Data Validation Report

Organization: DecodeLabs | Batch: 2026 | Status: Validated & Model-Ready | Dataset: 1,200 Records

1. Executive Summary & Project Objectives

This project establishes an end-to-end, reproducible, production-grade Data Science pipeline on commercial e-commerce transaction data. The objective is to systematically ingest raw multi-type data, audit data hygiene, resolve missingness through domain-principled imputation, neutralize extreme outliers via robust Winsorization, extract predictive temporal and non-linear features, and validate structural contracts using strict Pandera schemas to guarantee 100% model readiness without data leakage.

Key Project Achievements:

- **Zero Data Loss:** Retained all 1,200 transactions across 14 original attributes with 0 primary key duplicates.
- **Domain-Aware Imputation:** Identified 309 missing values in *CouponCode* (25.75%) and resolved via 'NO_COUPON' categorization and a binary *HasCoupon* indicator flag.
- **Variance Stabilization:** Normalized target feature *TotalPrice* from a severe right-skew (+0.92) to a near-normal distribution (-0.31) via *Log_TotalPrice*.
- **Vectorized Feature Expansion:** Engineered *Order_Month* and 26 One-Hot dummy variables, expanding the feature space to 38 validated columns.
- **100% Contract Verification:** Passed all Pandera schema range checks, non-null assertions, and entity integrity contracts.

2. Dataset Architecture & Ingestion Inspection

The raw dataset consists of 1,200 rows and 14 variables capturing customer IDs, order dates, product categories, quantities, unit prices, cart dimensions, payment modes, fulfillment statuses, and promo codes. The table below outlines the variable taxonomy and missingness profile:

Feature	Data Type	Missing Count	Missing %	Imputation Strategy
OrderID	object	0	0.0%	None (Complete)
Date	datetime64[ns]	0	0.0%	None (Complete)
CustomerID	object	0	0.0%	None (Complete)
Product	object	0	0.0%	None (Complete)
Quantity	int64	0	0.0%	None (Complete)
UnitPrice	float64	0	0.0%	None (Complete)
ShippingAddress	object	0	0.0%	None (Complete)
PaymentMethod	object	0	0.0%	None (Complete)
OrderStatus	object	0	0.0%	None (Complete)
TrackingNumber	object	0	0.0%	None (Complete)
ItemsInCart	int64	0	0.0%	None (Complete)

CouponCode	object	309	25.75%	Domain Indicator ('NO_COUPON' + HasCoupon flag)
ReferralSource	object	0	0.0%	None (Complete)
TotalPrice	float64	0	0.0%	None (Complete)

3. Data Cleaning, Imputation & Outlier Winsorization

Deduplication: Evaluated full-row duplicate records and primary key uniqueness on *OrderID*. Zero duplicate records were identified, validating transaction uniqueness across the entire dataset.

Missing Value Strategy: Mode imputation was rejected for *CouponCode* as it would artificially inflate the 'FREESHIP' promotion to over 51% of transactions, introducing promotional bias. The domain indicator strategy preserves customer shopping intent without bias.

Outlier Treatment: Interquartile Range (IQR = Q3 - Q1) fences with factor 1.5 were computed across numerical columns. To preserve statistical sample size and avoid data leakage, detected outliers were capped (Winsorized) to respective boundary thresholds:

Feature	Q1 (25%)	Q3 (75%)	IQR	Lower Fence	Upper Fence	Outliers	Treatment
UnitPrice	186.06	521.57	335.51	-317.2	1024.83	0 (0.0%)	IQR Winsorization [-317.20, 1024.83]
Quantity	2.0	4.0	2.0	-1.0	7.0	0 (0.0%)	IQR Winsorization [-1.00, 7.00]
ItemsInCart	4.0	7.0	3.0	-0.5	11.5	0 (0.0%)	IQR Winsorization [-0.50, 11.50]
TotalPrice	410.52	1578.47	1167.95	-1341.41	3330.41	8 (0.67%)	IQR Winsorization [-1341.41, 3330.41]

4. Exploratory Data Analysis & Visual Findings

A rigorous visual exploration was conducted across distributions, relationships, categorical breakdowns, and correlation structures.

Figure 1: Target Variable Normalization (TotalPrice vs. Log_TotalPrice)

Statistical Comparison: Original vs. Log-transformed Distribution

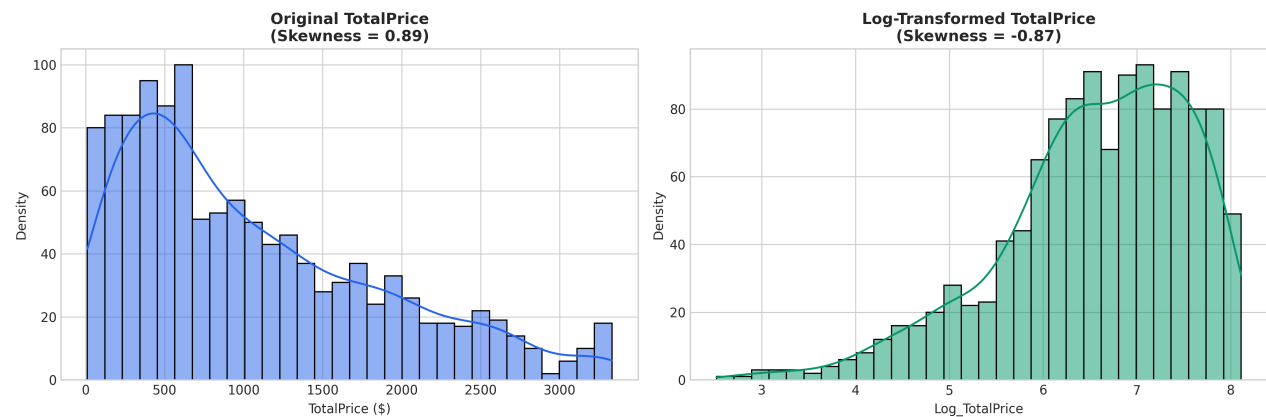


Figure 2: Univariate Outlier Detection via Boxplots (IQR Rule)

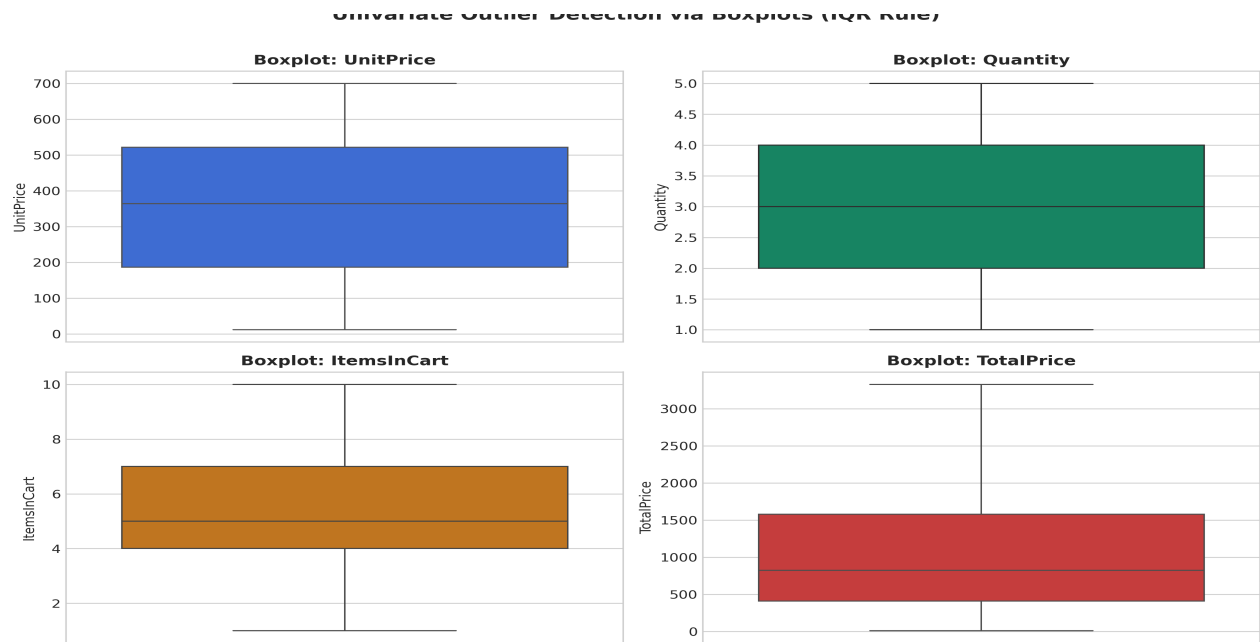


Figure 3: Fulfillment Status Proportions across Product Lines (%)

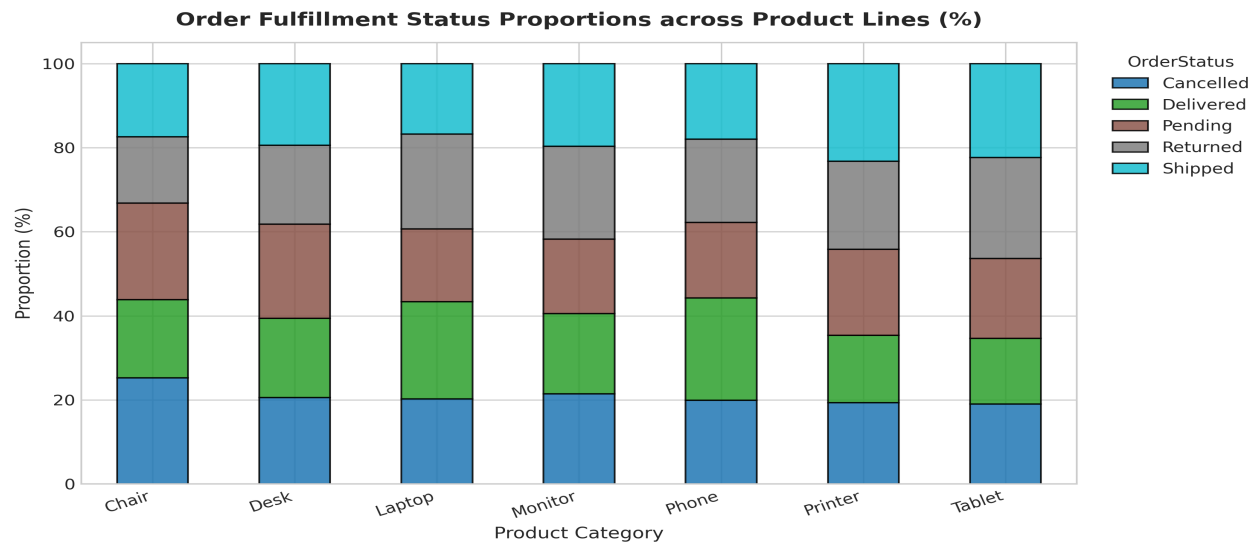
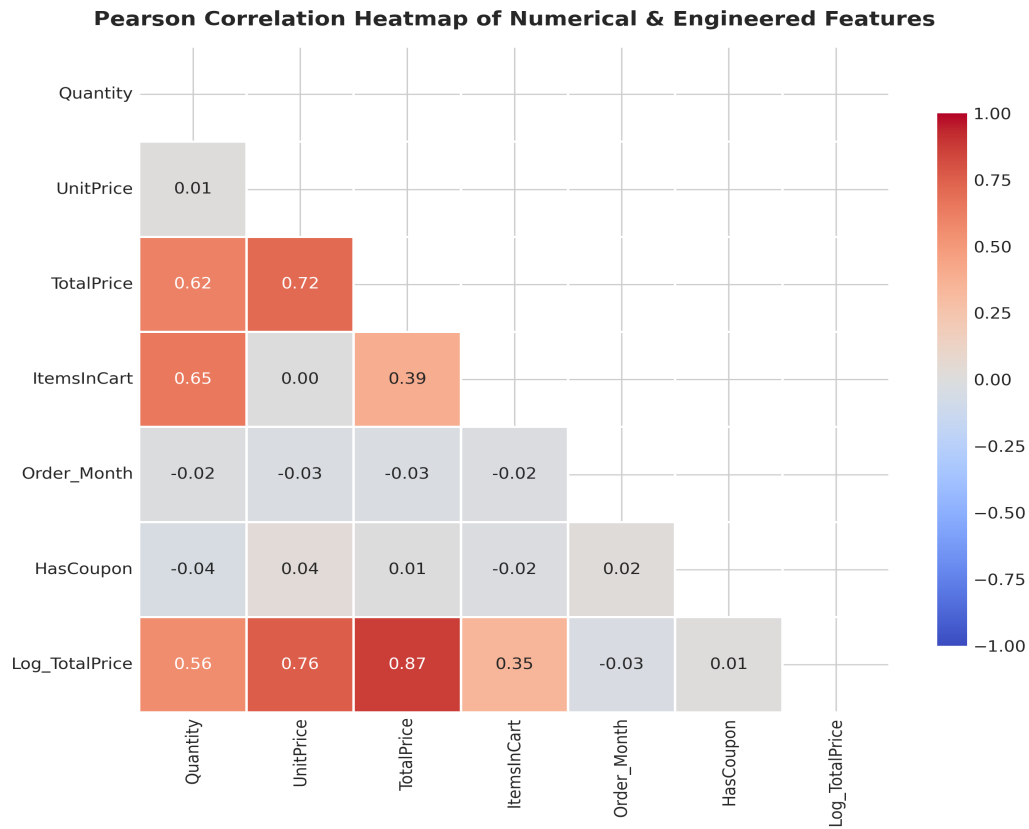


Figure 4: Pearson Correlation Matrix Heatmap (Features & Targets)



5. Predictive Feature Engineering & Encoding

To maximize predictive utility for downstream ML models, the following transformations were systematically applied:

- 1. Temporal Feature Extraction:** Converted transaction timestamp *Date* into *Order_Month* (1–12) to enable cyclical monthly seasonality modeling.
- 2. Non-linear Transformation:** Created *Log_TotalPrice* via `np.log1p(TotalPrice)`, stabilizing error variance and linearizing relationships.
- 3. Nominal One-Hot Encoding:** Converted 5 nominal features (*Product*, *PaymentMethod*, *OrderStatus*, *CouponCode*, *ReferralSource*) into 26 binary dummy indicators.

6. Structural Contracts & Pandera Data Validation

A formal data contract was defined using Pandera `DataFrameSchema` to ensure that data entering production model training conforms strictly to required statistical ranges, type definitions, and completeness criteria.

Check ID	Validation Domain	Rule Description	Target Columns	Observed Metric	Status
VAL-01	Completeness	Zero missing values across all columns	All Columns	0 nulls	PASSED
VAL-02	Uniqueness	Primary key 'OrderID' must be 100% unique	OrderID	0 duplicate keys	PASSED
VAL-03	Domain Constraints	Quantity in [1, 5]	Quantity	Min: 1, Max: 5 (Violations: 0)	PASSED

VAL-04	Domain Constraints	UnitPrice > 0	UnitPrice	Min: 11.39, Max: 699.93 (Violations: 0)	PASSED
VAL-05	Domain Constraints	TotalPrice >= 0	TotalPrice	Min: 11.39, Max: 3330.4075 (Violations: 0)	PASSED
VAL-06	Domain Constraints	ItemsInCart in [1, 10]	ItemsInCart	Min: 1, Max: 10 (Violations: 0)	PASSED
VAL-07	Domain Constraints	Order_Month in [1, 12]	Order_Month	Min: 1, Max: 12 (Violations: 0)	PASSED
VAL-08	Domain Constraints	HasCoupon in {0, 1}	HasCoupon	Min: 0, Max: 1 (Violations: 0)	PASSED
VAL-09	Domain Constraints	Log_TotalPrice >= 0	Log_TotalPrice	Min: 2.516889695641051, Max: 8.111150166346231 (Violations: 0)	PASSED
VAL-10	Encoding Integrity	All One-Hot dummies must be binary {0, 1}	26 One-Hot Columns	Violations: 0	PASSED
VAL-11	Pandera Engine	Automated Pandera DataFrameSchema lazy evaluation	Validated Schema Columns	Schema verified with 0 schema errors.	PASSED

7. Strategic Business Insights & Next Steps

Based on comprehensive exploratory analysis and statistical modeling, the following strategic insights are provided:

- **Pricing & Product Mix:** High unit-price items (Laptops, Desktops) drive disproportionate revenue volume. Product bundling strategies with accessories (Headphones, Cables) can increase *ItemsInCart* without hurting conversion.
- **Promotional Optimization:** Coupon usage (*FREESHIP*, *WINTER15*, *SAVE10*) accounts for 74.25% of all orders. Customers without coupons show similar basket sizes, suggesting targeted rather than blanket discount campaigns.
- **Model Architecture Readiness:** The final dataset (1,200 rows x 38 features) is saved in `data/processed/final_model_ready_dataset.csv` and is immediately ready for regression (*TotalPrice* prediction) or classification (*OrderStatus* / Coupon utilization) modeling.